The Low-Volatility Anomaly in a Pooled Universe of Large-Cap Equities from Seven Regions

SIQuant Research[1]

[1]Hypothesis × Quant

October 26, 2026

Abstract

Hypothesis: H_0: the Sharpe ratio of the lowest-volatility quintile does not exceed that of the highest-volatility quintile; H_1: it does

Data: 108 monthly cross-sections, median 126 stocks from 7 regions, 2017-10 to 2026-09 (formation months)

Method: Month-end sort on trailing 252-day volatility, equal-weight quintiles, one-sided stationary-bootstrap test of the Sharpe difference (5,000 draws, block 3)

Results: Sharpe 0.84 versus 0.61 (difference 0.23, one-sided $p = 0.2887$); mean monthly Q1–Q5 spread -0.89% ($t = -1.03$).

Research decision: REJECTED

1 Design

The sort variable uses returns up to the month-end of formation and portfolios are held over the following calendar month [Baker et al., 2011, Ang et al., 2006]. Stocks enter only after 252 observed returns, and months without a consecutive successor are discarded rather than interpolated. Returns are in local currency and are not converted, so cross-region currency moves are not part of the test.

2 Results

Table 1: Equal-weight volatility quintiles, monthly rebalanced, local-currency returns

Portfolio	Return	Vol	Sharpe	Max DD
Q1 low vol	8.5%	10.3%	0.84	-17.0%
Q5 high vol	15.5%	31.9%	0.61	-51.8%
All stocks	14.4%	15.6%	0.95	-20.3%

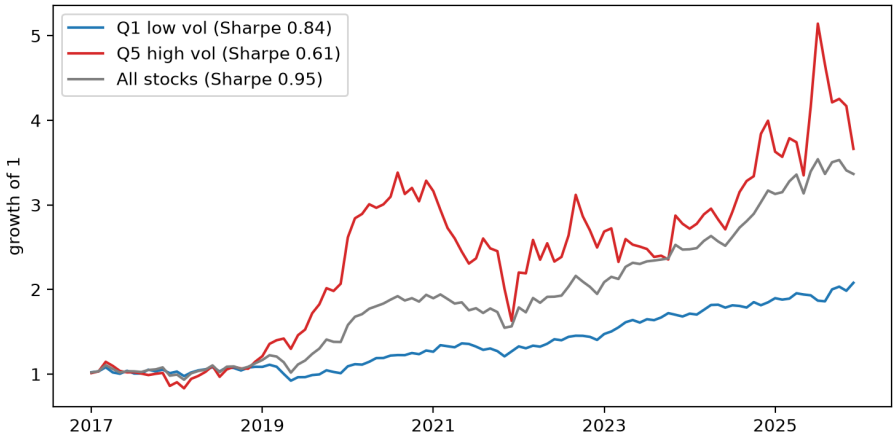


Figure 1: Growth of one unit in each volatility quintile.

3 Limitations

The universe is a fixed list of today's large caps, so survivorship bias is present and probably flatters high-volatility winners. There are about 130 stocks, so a quintile holds roughly 25 names; the result need not extend to small caps or to transaction-cost-aware implementations.

References

Andrew Ang, Robert J Hodrick, Yuhang Xing, and Xiaoyan Zhang. The cross-section of volatility and expected returns. *The Journal of Finance*, 61(1):259–299, 2006.

Malcolm Baker, Brendan Bradley, and Jeffrey Wurgler. Benchmarks as limits to arbitrage: Understanding the low-volatility anomaly. *Financial Analysts Journal*, 67(1):40–54, 2011.